

Relation Algebra

§8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

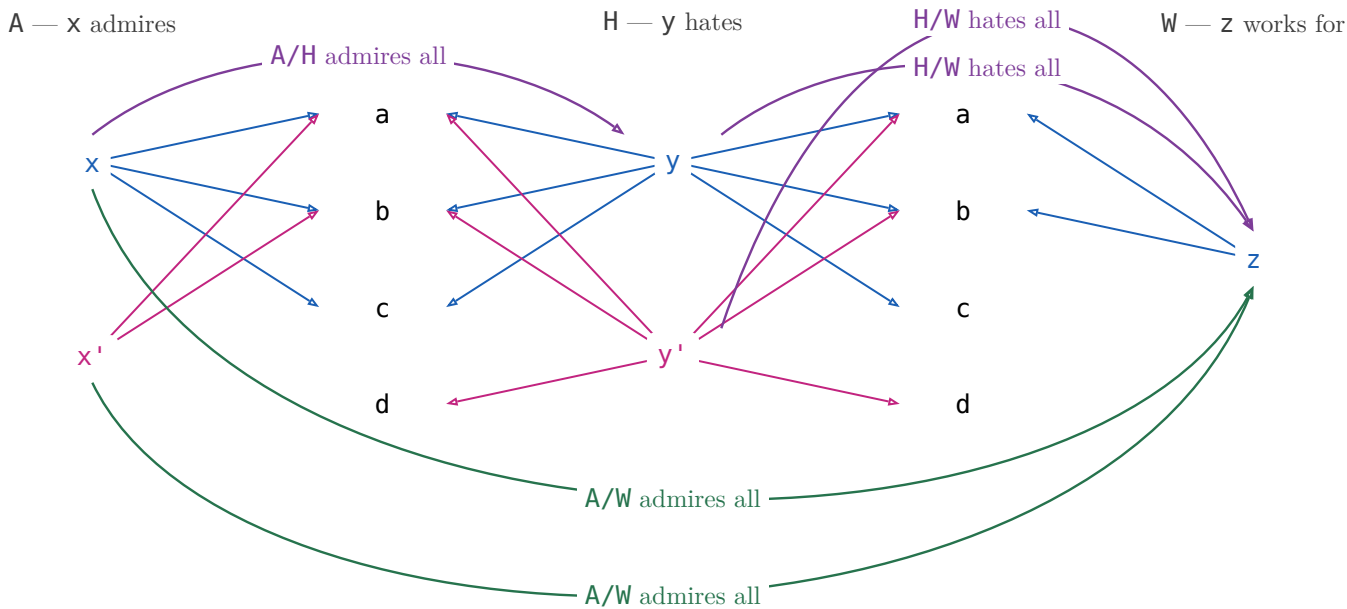
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§8.1. $(R/S)(S/W) \subseteq R/W$

(8.1a)



(8.1b)

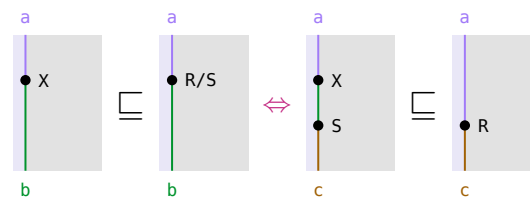
$x (A/H) y$ — x admires everyone y hates
 $y (H/W) z$ — y hates everyone z works for
 $x (A/W) z$ — x admires everyone z works for

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \subseteq R/W$.

(8.1c)

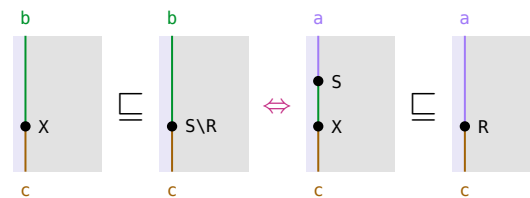
$$1 \quad X \in R/S \iff X \in S \subseteq R$$

X is any X -to- y pairing; one that only pairs X with a y such that X admires everyone y hates lies inside R/S , and R/S is the largest such.



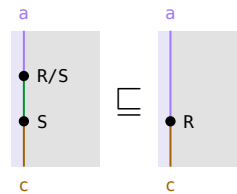
$$2 \quad X \in S \setminus R \iff X \in S \subseteq R$$

The mirror — divide on the left when X comes first.



$$3 \quad (R/S) \subseteq R$$

There is a y such that X admires everyone y hates, and p is one of the people y hates — then X admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.

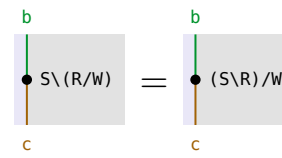


4	$S (S \setminus R) \sqsubseteq R$ The mirror.	
5	associate: $R / (S_1 S_2) = (R / S_2) / S_1$ A friend's enemy is two hops: divide by the far end first.	
6	$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
7	maps: $f (R/S) = (fR) / S$ Rename x before or after dividing — the licence to write fR/S .	
8	$R / (fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
9	$(R/S) (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
10	$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
11	$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
12	$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
13	$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
14	$R / (S_1 \cup S_2) = R / S_1 \cap R / S_2$ Admiring a combined hate-set is admiring each set in full.	

15

$$S \setminus (R/W) = (S \setminus R) / W$$

Which is why $S \setminus R/W$ needs no bracket.



Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.